

LightSense - Long Distance

Uwe Rieger
Yinan Liu
u.rieger@auckland.ac.nz
yinan.liu@auckland.ac.nz
arc/sec Lab
University of Auckland
Auckland, New Zealand

Tharindu Kaluarachchi
tikzoxs@gmail.com
Augmented Human Lab
University of Auckland
Auckland, New Zealand

Amit Barde
Huidong Bai
Alaeddin Nassani
amit.barde@auckland.ac.nz
huidong.bai@auckland.ac.nz
alaeddin.nassani@auckland.ac.nz
Empathic Computing Lab
University of Auckland
Auckland, New Zealand

Suranga Nanayakkara
suranga@ahlab.org
Augmented Human Lab
National University of Singapore
Singapore

Mark Billinghamurst
mark.billinghurst@auckland.ac.nz
Empathic Computing Lab
University of Auckland
Auckland, New Zealand



ABSTRACT

'LightSense - Long Distance' explores remote interaction with architectural space. It is a virtual extension of the project 'LightSense,' which is currently presented at the exhibition 'Cyber Physical: Architecture in Real Time' at EPFL Pavilions in Switzerland. Using numerous VR headsets, the setup at the Art Gallery at SIGGRAPH Asia establishes a direct connection between both exhibition sites in Sydney and Lausanne.

'LightSense' at EPFL Pavilions is an immersive installation that allows the audience to engage in intimate interaction with a living architectural body. It consists of a 12-meter-long construction that combines a lightweight structure with projected 3D holographic animations. At its core sits a neural network, which has been trained on sixty thousand poems. This allows the structure to engage, lead, and sustain conversations with the visitor. Its responses are truly associative, unpredictable, meaningful, magical, and deeply emotional. Analysing the emotional tenor of the conversation, 'LightSense' can transform into a series of hybrid architectural volumes, immersing the visitors in Pavilions of Love, Anger, Curiosity, and Joy.

'LightSense's' physical construction is linked to a digital twin. Movement, holographic animations, sound, and text responses are controlled by the cloud-based AI system. This combination creates a location-independent cyber-physical system. As such, the 'Long

Distance' version, which premiered at SIGGRAPH Asia, enables the visitors in Sydney to directly engage with the physical setup in Lausanne. Using VR headsets with a new 360-degree 4K live streaming system, the visitors find themselves teleported to face 'LightSense', able to engage in a direct conversation with the structure on-site.

'LightSense - Long Distance' leaves behind the notion of architecture being a place-bound and static environment. Instead, it points toward the next generation of responsive buildings that transcend space, are capable of dynamic behaviour, and able to accompany their visitors as creative partners.

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